



Spatial availability of federally qualified health centers and disparities in health services utilization in medically underserved areas

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ABSTRACT

Federally qualified health centers (FQHCs) improve access to care for important health services (e.g., preventive care), particularly among marginalized and underserved communities. However, whether spatial availability of FQHCs influences care-seeking behavior for medically underserved residents is unclear. The objective of this study was to examine the relationships of present-day zip-code level availability of FQHCs, historic redlining, and health services utilization (i.e., at FQHCs and any health clinic/facility) in six large states. We further examined these associations by states, FQHC availability (i.e., 1, 2–4 and ≥ 5 FQHC sites per zip code) and geographic areas (i.e., urbanized vs. rural, redlined vs. non-redlined sections of urban areas). Using Poisson and multivariate regression models, we found that in medically underserved areas, having at least one FQHC site was associated with greater likelihood of patients seeking health services at FQHCs [rate ratio (RR) = 3.27, 95%CI: 2.27–4.70] than areas with no FQHCs available, varying across states (RRs = 1.12 to 6.33). Relationships were stronger in zip codes with ≥ 5 FQHC sites, small towns, metropolitan areas, and redlined sections of urban areas (HOLC D-grade vs. C-grade: RR = 1.24, 95%CI: 1.21–1.27). However, these relationships did not remain true for routine care visits at any health clinic or facility ($\beta = -0.122$; $p = 0.008$) or with worsening HOLC grades ($\beta = -0.082$; $p = 0.750$), potentially due to the contextual factors associated with FQHC locations. Findings suggest that efforts to expand FQHCs may be most impactful for medically underserved residents living in small towns, metropolitan areas and redlined sections of urban areas. Because FQHCs can provide high quality, culturally competent, cost-effective access to important primary care, behavioral health, and enabling services that uniquely benefit low-income and marginalized patient populations, particularly those who have been historically denied access to health care, improving availability of FQHCs may be an important mechanism for improving health care access and reducing subsequent inequities for these underserved groups.

1. Introduction

Disparities in access to quality preventive health care services in racially and socioeconomically marginalized communities remain significant drivers of health inequities in the United States (U.S.) (Howell et al., 2008, 2016; Kothari et al., 2016; Lucas et al., 2006; Parker, 2021). Several studies have elucidated that health outcomes in marginalized communities may in part be attributed to residential segregation and disparities in health care quality (Howell et al., 2016; Ko and Ponce,

2013; LaVeist et al., 2009; Whittle et al., 1993). Additionally, a variety of state and neighborhood-level factors can influence health inequities in marginalized communities. For example, state-administered health care policies (e.g., Medicaid expansion), built environments (e.g., access to green space, healthy food stores, residential proximity to industrial facilities or major highways, availability of social services), and social structures (e.g., racial segregation, policing, criminalization of poverty) may contribute to disparities in access to preventive services (e.g., primary care visits, dental services, medications for diabetes), and worsen

Abbreviations: β , beta coefficient; FQHC, Federally Qualified Health Centers; HOLC, Home Owners' Loan Corporation; RR, Rate Ratio.

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health inequities in marginalized communities (Cole et al., 2017a; Edward and Biddle, 2017; Fisher et al., 2000; Ncube et al., 2016; Ross and Wu, 1995; Ryvicker and Sridharan, 2018; Wennberg and Cooper, 1996).

Federally qualified health centers (FQHCs) are federally funded community-based health care facilities that primarily serve marginalized communities and provide “comprehensive, culturally competent, high-quality primary care services” to vulnerable individuals (Shi et al., 2012; Health Resources & Services Administration, 2016). FQHCs serve approximately 30 million people across the U.S. In 2019, more than 91% of patients using health care services at FQHCs reported incomes below 200% of the federal poverty level (i.e., \$27,180 for a single person and \$55,500 for a household of four), 48% were enrolled in Medicaid, and 23% were uninsured (Health Resources & Services Administration, 2021a). Additionally, 63% of patients receiving care at FQHCs were from racial/ethnic minoritized groups, 37% being Hispanic/Latino and 22% African American. Prior studies have shown that availability of FQHCs can improve access to important preventive health services and improve care quality for underserved populations (e.g., Medicaid-enrollees) (Cole et al., 2017b; Hadley and Cunningham, 2004; Knight, 2011; Mehta et al., 2016; Shin et al., 2013). Furthermore, more recent studies have examined the impact of spatial expansion of FQHCs before and after the implementation of Affordable Care Act (Behr et al., 2022; Chang et al., 2019). To date, however, no research has investigated whether differences exist across states in terms of the impact of spatial availability to FQHCs on patients’ utilization of health services in particular geographic areas (i.e., rural, small towns or specific sections within metropolitan areas) that are deemed medically underserved.

Federal regulations in part dictate where FQHCs can be located – they must be located in medically underserved areas or in areas with medically underserved populations (MUAs) (Salinsky, 2010). MUAs are defined by Health Resources and Services Administration (HRSA) as areas having too few primary care providers, high infant mortality rates, high poverty rates, or a large elderly population (Health Resources and Services Administration, 2022). Yet, there are many MUAs that do not have any FQHCs, which may be driven by the lack of commitment to state-level and health system-level investment in FQHC systems. Meanwhile, billions of federal dollars have been made available through the Affordable Care Act’s Community Health Center Fund and the American Rescue Plan to build, expand, and support FQHCs across the

U.S. (H.R.133, 2020; Commonwealth of Massachusetts, 2021; Health Resources & Services Administration, 2021b; Kaiser Family Foundation, 2019; National Association of Community Health Centers, 2016; National Association of Community Health Centers, 2021). This provides unique opportunities for geographic expansion of FQHCs across different states.

Existing inequities in access to health care services are driven, in part, by historic and discriminatory housing, lending, and urban planning practices in the U.S., known as redlining – a racialized zoning practice implemented in the 1930s and 1940s (Krieger et al., 2020b; Nardone et al., 2020; Nelson et al., 2019). Redlining is an operationalized and measurable way of quantifying racial segregation and structural racism, essentially serving as a proxy indicator of racialization, discrimination, and neighborhood disadvantage (i.e., fewer health care facilities, resources, and funding) observed in certain U.S. urban areas (Hillier, 2003, 2005; Lee et al., 2022; Michney, 2022; Swope et al., 2022) (Fig. 1).

Prior to the establishment of FQHCs in the 1980s, the persistent segregation caused by structural factors like redlining has contributed to present-day inequities in access to health care among residents who have historically faced barriers and were denied access to quality and trusted health care services (Politzer et al., 2001; Rojas Marquez et al., 2023). Despite recent efforts to allocate FQHCs in MUAs, recent research continues to illuminate worse access to health care services among current residents of historically redlined areas, compared to those living in non-redlined areas (Supplementary Fig. 1). For example, recent studies report that residential redlining and disparities in access to health care are linked with present-day delays in disease diagnosis, annual physical exams, treatment of health conditions, decreased behavioral health clinicians available (e.g., psychologists, counselors, and therapists) (Erikson et al., 2022; Krieger et al., 2020b; Motairek et al., 2022) and larger Medicaid or uninsured populations (Allard et al., 2020; Nardone et al., 2020). Further, Krieger et al. (2020b) and Nardone et al. (2020) have shown that residents living in redlined neighborhoods had protective effects on certain types of cancer, potentially due to the lack of access to health services or health insurance, leading to missed cancer diagnosis (Gaskin et al., 2012a, 2012b; Jackson et al., 2016), which provides an opportunity to intervene through the FQHC system.

Collectively, current evidence indicates existing disparities in access and quality of care and potential opportunities for ameliorating health

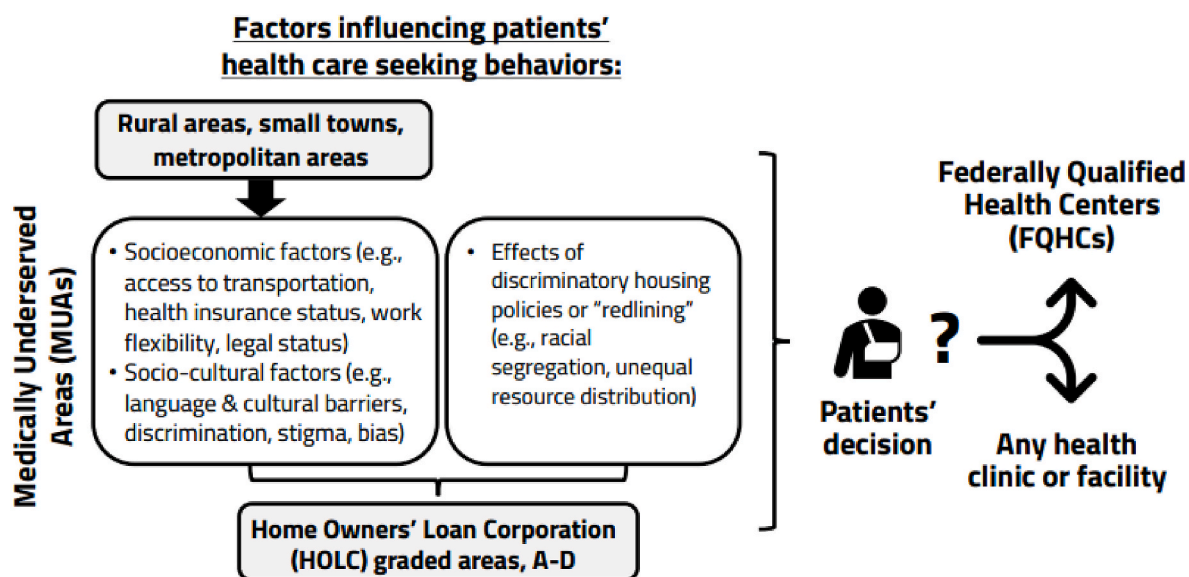


Fig. 1. Conceptual model depicting factors influencing patients’ health care seeking behaviors to either (a) federally qualified health centers (FQHCs) or (b) any health clinic or facility in distinct geographic areas (i.e., rural areas, small towns, metropolitan areas and Home Owners’ Loan Corporation (HOLC) graded areas) in areas designated medically underserved.

inequities. However, our understanding of how policies structuring neighborhood composition and resources influence spatial access to health care across states, specifically FQHCs, is lacking and significant knowledge gaps remain. First, although FQHCs play an important role in supporting low-income populations' access to important health services (e.g., preventive care, chronic disease management), it remains unknown whether: (a) FQHCs are adequately available in medically underserved areas and historically redlined areas, and (b) FQHCs are being utilized by medically underserved populations. Second, mechanisms by which neighborhood environments (e.g., neighborhood socioeconomic status, racial composition, access to transportation) across different states play a role in FQHC utilization merits further investigation.

To address current knowledge gaps, we sought to: (1) spatially examine the relationships between present-day availability of FQHCs and patients' utilization of health care services using national administrative data as well as spatial and Census data in medically underserved areas in six large states and assess indicators of health care utilization; and (2) assess whether historic redlining influences FQHC locations.

2. Methods

2.1. Study design and areas

Using 2019 national administrative data, Centers for Disease Control and Prevention (CDC) PLACES database, and digitized Home Owners' Loan Corporation (HOLC) appraisal maps created in 1938, we conducted cross-sectional analyses to examine the associations between present-day zip code-level availability of FQHCs and patients' utilization of health care services (i.e., at FQHCs or at any other health clinic/facility) in six U.S. states. We also examined whether health care utilization varied by state, FQHC availability, geographic areas (i.e., urbanized and rural areas) and HOLC-graded areas (i.e., A (best) – D (worst) graded areas).

We included six states for our analyses: California (CA), Florida (FL), Illinois (IL), Massachusetts (MA), New York (NY), and Texas (TX). Out of 4598 FQHC delivery sites available across six states, our analyses were based on 2251 FQHC delivery sites that were located in MUAs. These states were chosen based on two criteria: first, that sufficient data were available for analyses (i.e., having enough FQHC delivery sites in both urbanized and rural areas); and second, that they represented diversity in geography by US census region (i.e., Northeast, Midwest, South or West), population demographics, neighborhood environments, and health policy environments (e.g., smoking bans, flextime for physical activity or attendance of health programs, Medicaid expansion).

Out of 8407 zip codes across six states, we limited this study to 1343 zip codes located in MUAs for two reasons: first, to account for baseline differences in sociodemographic and neighborhood characteristics of FQHC patients compared to the rest of the state; and second, since federal regulations in part dictate where FQHCs can be located (i.e., MUAs), results from this study may help policymakers determine future FQHC locations. Since HRSA's data on MUAs are available in different spatial units (e.g., county, subcounty, census tracts, zip code levels), all MUAs were spatially aligned to the zip code-level by spatially joining MUA polygons to the zip code boundaries (Health Resources and Services Administration, 2022). We excluded zip codes that did not fall within MUAs. Spatial joints and maps were developed using ArcGIS (version 10.4, ESRI, Redlands, CA).

2.2. Outcome assessment

We examined two study outcomes. First, we examined the total number of FQHC patients (or patient volume) per zip code, inclusive of all ages, who utilized FQHC health services from any FQHC setting (i.e., hospital/clinic, school, mobile van, nursing home, correctional facility, and domestic violence center) in 2019. Data on FQHC patients were

obtained from the HRSA 2019 Uniform Data System (UDS) (Health Resources and Services Administration, 2019). Zip codes were based on the FQHCs that patients used for health care services, instead of zip codes of patients' residences. The UDS includes information from all HRSA-funded FQHCs and collects annual data on patient characteristics, organizational and structural characteristics, quality of care, and service volume. All UDS data are reported at the FQHC-level, capturing the location of facilities including when FQHCs have more than one site but reporting limited delivery site information.

Second, we examined zip code-level prevalence estimates of annual visit to doctor for routine check-up among adults ages 18 years or older at any hospital/clinic, including non HRSA-funded settings. The zip code-level prevalence data were obtained from the CDC PLACES database, which includes information from the 2019 Behavioral Risk Factor Surveillance System (BRFSS), the 2010 U.S. Census population, and the American Community Survey estimates (CDC, 2021). The prevalence estimates are self-reported from respondents that were randomly chosen from across the country. CDC PLACES provides the prevalence estimates of health indicators based on zip codes of respondents' residences by using multilevel regression and post-stratification approach. We also obtained information about the population density for each zip code from the U.S. Census to adjust for in the statistical analyses (US Census, 2020). Boston University's Institutional Review Board (IRB) deemed the study exempt and waived informed consent because all data used in this study are publicly available.

2.3. Exposure assessment

To examine the relationships of present-day zip code-level availability of FQHCs, historic redlining, and health services utilization, we used two exposure measurements (1) FQHC availability (i.e., zip codes that had 0 FQHCs vs. zip codes that had 1 or more FQHCs) (Section 2.3.1); and (2) redlining status (i.e., comparison of HOLC-graded areas) (Section 2.3.2).

2.3.1. Federally qualified health centers

We measured FQHC availability based on the number of FQHC delivery sites available by zip code. Exposed areas were defined as zip codes having ≥ 1 FQHC sites, whereas unexposed areas were defined as zip codes within MUAs with no available FQHC sites. Among zip codes deemed MUAs, we matched exposed areas with unexposed areas by first creating eight-mile buffers surrounding each FQHC site based on site addresses, and exclusively including unexposed areas or zip codes that had no FQHCs and did not overlap with the eight-mile FQHC buffers (Supplementary Fig. 2). This distance was chosen because eight miles is the average distance patients travel to access emergency care in the U.S., providing a proxy measure for overall access to health care (Tolpadi et al., 2022). Selecting unexposed areas without any nearby FQHCs (or within an 8-mile radius) reduces the potential for exposure misclassification due to FQHC exposure.

For linking zip code-level outcome data and FQHC availability, information on FQHCs' patient volume and prevalence estimates for routine care visits in the past twelve months were available at the zip code-level for year 2019 and were linked with the exposure variable available at the zip-code level. We further examined exposed areas based on FQHC availability distributions: Low-availability or 1 FQHC site per zip code, medium-availability or 2–4 FQHC sites per zip code, and high-availability or ≥ 5 FQHC sites per zip code.

2.3.2. Historically redlined neighborhoods

We obtained census tract-level HOLC-scored data from the University of Richmond Mapping Inequality digitized maps (Nelson et al., 2019). We evaluated the associations between redlining and FQHC utilization, focusing on major U.S. cities across the six states (CA, FL, IL, MA, NY, TX) that were scored based on HOLC's set of criteria (Supplementary Table 1). Our aim was to document the how FQHC use patterns

may differ in zip codes based on local FQHC presence and HOLC status for two main reasons: (1) FQHCs may provide an opportunity to facilitate access to important health services for medically underserved populations in areas (i.e., redlined areas) that have been historically denied access to health care (Fig. 1); and (2) Krieger et al. (2020b) and Nardone et al. (2020) reported protective effects of certain types of cancer among residents living in redlined neighborhoods and this could potentially be related, in part to reduced health care access in these areas. The color-coded areas were scored in the 1930s and 1940s, in over 239 U.S. cities, based on the potential loan lending risks and defaults in mortgage repayments, neighborhood racial/ethnic composition, and neighborhood environments (i.e., presence of industrial facilities) and graded as follows: A (“best” or green), B (“desirable” or blue), C (“declining” or yellow) and D (“hazardous” or red).

To link zip code-level outcome data and redlining status, we followed the approach used by Krieger et al. (2020a). We first spatially joined 1938 HOLC-graded census tracts and 2019 census tract shapefiles using ArcGIS spatial join feature; we only included the census tracts that overlapped. Each census tract was assigned a HOLC-grade (Supplementary Fig. 3). Then, we overlaid zip code boundaries to the spatially joined HOLC-graded census tracts and designated each zip code a HOLC grade; at least 50% of the land in that zip code needed to have the assigned HOLC grade. For “mixed” zip codes (i.e., zip codes containing two or more HOLC-graded census tracts), we assigned a grade based on the proportion of HOLC-graded census tract that predominantly covered the zip code land and had at least 50% of the zip code land coverage. For the analyses, we compared adjacent HOLC-graded areas (B vs. A, C vs. B, D vs. C, A-D) and across HOLC grades (A through D, as numerics with A = 1 and D = 4) among zip codes that had at least one FQHC to better isolate the impact of redlining, and further stratified by state.

2.4. Covariates

Covariates in this study included zip code-level data on: total population; proportion of residents who identify as Black/African-American and Hispanic; non-U.S. citizens; proportion of homes owning a vehicle; proportion of unemployed residents (ages 16 and older); proportion of homes without a computer; proportion of homes receiving Supplemental Nutrition Assistance Program (SNAP) benefits; and proportion of residents who had less than high school education (Supplementary Table 2). These covariates were chosen initially based on evidence in the existing literature on well-established factors known to associate with health services utilization (National Academies of Sciences and Medicine, 2018), as well as by consulting subject experts in health services and policy. We used available zip code data (2019) from the American Community Survey (US Census, 2020). Information regarding urban and rural areas were obtained from the United States Department of Agriculture (USDA) databases (USDA, 2020). USDA classified urbanized and rural areas into four categories, using measures of population density, urbanization, and daily commuting by zip code. Metropolitan areas are defined as urbanized areas with a population of $\geq 50,000$, micropolitan areas as urbanized areas with a population of 10,000 to 49,999, small towns as urbanized areas with a population of 2500 to 9,999, and rural areas as rural areas with a population of less than 2500.

2.5. Statistical analysis

We conducted descriptive analyses using variance of analysis (ANOVA), t-tests, or chi-square tests, as appropriate, to determine whether there were statistically significant differences between exposed (areas with ≥ 1 FQHC site) vs. unexposed areas (areas with no FQHC sites). Using zip code as the unit of analysis, we examined the relationships between FQHC exposures and health care utilization using two methods.

First, we used Poisson regression models to regress the total number of patients per zip code who utilized FQHCs by dichotomized exposures:

(1) zip codes based on FQHC availability (0 vs. ≥ 1 FQHC sites per zip code); and (2) zip codes based on HOLC-graded designations. We conducted stratified analyses by state, FQHC availability, geographic areas, and HOLC-grades. We adjusted for the relevant covariates in the models (as listed in Section 2.4) and the total population by including information on 2019 population density from the U.S. Census as an offset variable. We ran a total of 35 models: seven models for states (i.e., one model for six states and each model per state), three models for FQHC availability (i.e., low, medium, and high FQHC availability), four models by geographic areas (i.e., metropolitan, micropolitan, small towns, and rural areas), and 21 models for HOLC grade comparisons by state (i.e., B vs. A, C vs. B and D vs. C for each state (18 models) and all 6 states (3 models)) (USDA, 2020). We obtained the estimated rate ratios (RRs) of FQHC patient volume from the models and respective 95% confidence intervals. The all subsets model building approach was used to select the models by assessing the 10% differences in the effect estimates from the full model using backward selections.

To examine whether availability of FQHCs was associated with the prevalence of annual routine care visits at any health clinic or facility, we conducted multivariate linear regression analyses, adjusting for covariates, as listed in Section 2.4. We also conducted stratified analyses by state, geographic areas, and HOLC-grades. We ran a total of 12 models for each category: seven models for states (i.e., one model for all six states and each model per state), four models by geographic areas (i.e., metropolitan, micropolitan, small towns and rural areas) and one model for HOLC graded designations (i.e., A through D, where we converted HOLC graded letters into numerics; A = 1, B = 2, C = 3, D = 4). We obtained beta coefficient values (β) and standard errors (SE) for each model. We assessed goodness of fit using backward elimination process, for which less difference in p-values from the main model indicates a better fit. We used two-tailed $\alpha = 0.05$ to determine statistical significance and all statistical analyses were performed using SAS software version 9.4 (SAS, 2015).

3. Results

3.1. Study characteristics and sociodemographic data by FQHC availability

Table 1 provides the characteristics of medically underserved zip codes and study populations by exposure (exposed vs. unexposed zip codes) based on FQHC locations in six states. The sample represents a total of 5,348,189 patients in 2251 FQHC sites in six states. The average population in FQHC zip codes was 9.9 times higher compared to non-FQHC zip codes. The weighted FQHC patient volume (weighted by population) observed in zip codes with at least one FQHC available was nearly four times the number of patients in zip codes with no FQHC available. The average prevalence of annual routine care visits in zip codes with at least one FQHCs was 73.5 ± 5.2 , whereas the average prevalence estimates in zip codes with no available FQHCs was 76.1 ± 3.3 .

We observed notable differences in FQHC availability in metropolitan areas compared to rural areas, where metropolitan areas had 13 times higher number of zip codes with at least one FQHC available than in rural areas and small towns. Health insurance coverage also varied across FQHC patients, with nearly half of the patients being Medicaid-enrollees and nearly one-fifth being uninsured. Regarding race and ethnicity, in zip codes with at least one FQHC site available, we observed that the percentage of White, Black, and Hispanic population were 2–3 times higher and nearly 10 times higher for Asian population, compared to zip codes that had no FQHC sites.

3.2. Variation in the associations by states and FQHC availability

Fig. 2 shows the distribution of FQHC delivery sites and MUAs (in black outlines) in six states (CA, FL, IL, MA, NY, TX). Across all six states,

Table 1

Characteristics of medically underserved zip codes based on federally qualified health center (FQHC) locations in six states (N = 2,251 FQHC delivery sites), 2019.

	Exposed areas ^a				Unexposed areas ^a
Characteristics	FQHCs available (≥ 1 FQHC sites)	Low availability (1 FQHC sites)	Medium availability (2–4 FQHC sites)	High availability (≥ 5 FQHC sites)	No FQHCs available (0 FQHC sites)
FQHC-level characteristics					
FQHC patient volume, No.	5,312,153	961,376	2,211,679	2,139,098	36,636
FQHC patient volume/ population	0.19	0.12	0.17	0.28	0.05
Prevalence of routine care visits, mean (SD)	73.5 (5.2)	73.5 (5.2)	72.9 (4.9)	72.2 (5.4)	76.1 (3.3)
Health insurance source, No. (%)					
Private	722,995 (13.6)	151,723 (15.8)	298,675 (13.5)	272,597 (12.7)	8,363 (22.8)
Medicare	402,400 (7.6)	73,070 (7.6)	167,031 (7.6)	162,299 (7.6)	4,297 (11.7)
Medicaid/CHIP	3,122,437 (58.8)	518,977 (54.0)	1,313,444 (59.4)	1,290,016 (60.3)	16,675 (45.5)
Uninsured	1,064,321 (20.2)	217,606 (22.6)	432,529 (19.6)	414,186 (19.4)	7,301 (19.9)
Zip code-level characteristics					
Total zip codes	749	292	315	142	230
Metropolitan areas ^b	646	220	287	139	78
Micropolitan areas ^c	52	31	19	2	45
Small towns ^d	30	23	6	1	58
Rural areas ^e	21	18	3	0	48
Total population ^f	28,076,277	7,770,825	12,778,194	7,527,258	726,560
% Race/ethnicity, mean (SD)					
White	60.9 (24.5)	70.7 (22.3)	57.8 (23.8)	47.7 (22.5)	90.5 (12.9)
Black or AA	15.4 (20.4)	12.5 (18.6)	15.9 (21.1)	20.3 (21.3)	3.2 (6.6)
AI/AN	0.7 (1.2)	0.8 (1.8)	0.6 (0.6)	0.7 (0.5)	1.4 (8.6)
Asian	7.0 (10.5)	4.8 (9.3)	8.6 (11.5)	8.2 (9.7)	0.6 (1.1)
Native Hawaiian or PI	0.2 (0.4)	0.1 (0.3)	0.2 (0.5)	0.2 (0.4)	0.1 (0.2)
Some other race	12.4 (13.0)	8.2 (10.8)	13.2 (12.9)	19.2 (14.0)	2.1 (5.1)
≥ 2 Races	3.4 (2.2)	3.0 (1.8)	3.7 (2.3)	3.7 (2.4)	2.1 (4.0)
Hispanic	38.9 (27.2)	34.9 (27.5)	38.5 (26.5)	48.1 (25.8)	9.7 (14.7)
% HH below poverty level, mean (SD)	18.9 (9.2)	17.0 (8.8)	18.8 (8.9)	22.8 (9.3)	13.3 (8.1)
% HH receiving SNAP, mean (SD)	17.7 (11.4)	15.9 (10.5)	17.3 (11.0)	22.2 (13.0)	11.3 (7.5)
% HH employed ≥ 16 , mean (SD)	57.8 (8.5)	56.6 (8.8)	58.6 (8.4)	58.5 (7.5)	52.9 (12.1)
% HH unemployed ≥ 16 , mean (SD)	6.9 (3.6)	6.5 (3.8)	6.8 (3.3)	8.0 (3.6)	5.2 (4.9)
% HH completed high school, mean (SD)	26.1 (8.3)	27.6 (8.6)	25.4 (8.2)	24.5 (7.7)	12.0 (7.4)
Total no. of vehicles owned per HH	8,400,555	2,710,430	3,832,080	1,858,045	260,110
HH vehicles/No. of HHs	0.87	1.01	0.84	0.63	0.93

Abbreviations: AA, African-American; AI/AN: American Indian or Alaskan Native; FQHC, Federally Qualified Health Centers; HH, household; NA, not applicable; PI: Pacific Islander; SD, standard deviation; SNAP, Supplemental Nutrition Assistance Program.

Note: In bold are statistically significant values, with $p < 0.001$, tested as dichotomous variables (0 FQHC sites vs. ≥ 1 FQHC sites) and as multichotomous variables (0 FQHC sites vs. 1, 2–4 and ≥ 5 FQHC sites).

^a The categories represent the number of FQHC sites per zip code.

^b Metropolitan areas: Urbanized areas with a population of $\geq 50,000$.

^c Micropolitan areas: Urbanized areas with a population of 10,000 to 49,999.

^d Small towns: Urban areas with a population of 2,500 to 9,999.

^e Rural areas: Rural areas with a population of less than 2,500.

^f American Community Survey (2019) for population and demographics.

the states located in the West (California) or South (Florida and Texas) had notably larger proportions of MUAs compared to the Northeast and Midwest states (New York, Illinois, and Massachusetts).

When compared to zip codes with no FQHCs, zip codes with at least one FQHC had higher likelihood of patients receiving care through an FQHC (RR: 3.27, 95%CI: 2.27–4.70) (Table 2). These associations varied significantly across states, with RRs ranging from 1.12 (Florida) to 6.33 (New York). When we examined the associations based on the number of FQHCs available by zip codes, the RR for zip codes with one FQHC site was 2.07 (95%CI: 1.86–2.30), 5.48 for zip codes with two to four FQHCs (95%CI: 3.64–8.25), and 6.38 for zip codes with at least five FQHCs (95%CI: 5.30–7.68).

Across six states, we found statistically significant decreases in the prevalence of annual routine care visits in zip codes with FQHC availability ($\beta = -0.122$; $p = 0.008$) (Table 3). These relationships varied by states, with beta coefficient values ranging from -0.307 (Florida) to 0.174 (Massachusetts), although these findings were not statistically

significant. Overall, in zip codes that had FQHCs available, we found decreases in the prevalence of routine care visits at any health clinic or facility in five states (i.e., California, Florida, Illinois, New York, and Texas), but increase in the prevalence of routine care visits in the state of Massachusetts ($\beta = 0.174$; $p = 0.148$).

3.3. Variation in the associations by geographic areas and neighborhoods

Zip codes with at least one FQHC were associated with greater likelihood of patients receiving health services through the FQHC system, particularly in small towns (RR: 6.07, 95%CI: 5.18–7.12), followed by metropolitan areas (RR: 4.51, 95%CI: 3.18–6.39), rural areas (RR: 2.07, 95%CI: 1.82–2.35) and micropolitan areas (RR: 1.65, 95%CI: 1.40–1.96) (Table 2).

In terms of the prevalence of routine care visits at any health clinic or facility, we also found positive associations with increasing availability of FQHCs in rural areas ($\beta = 0.269$; $p = 0.679$), although these findings

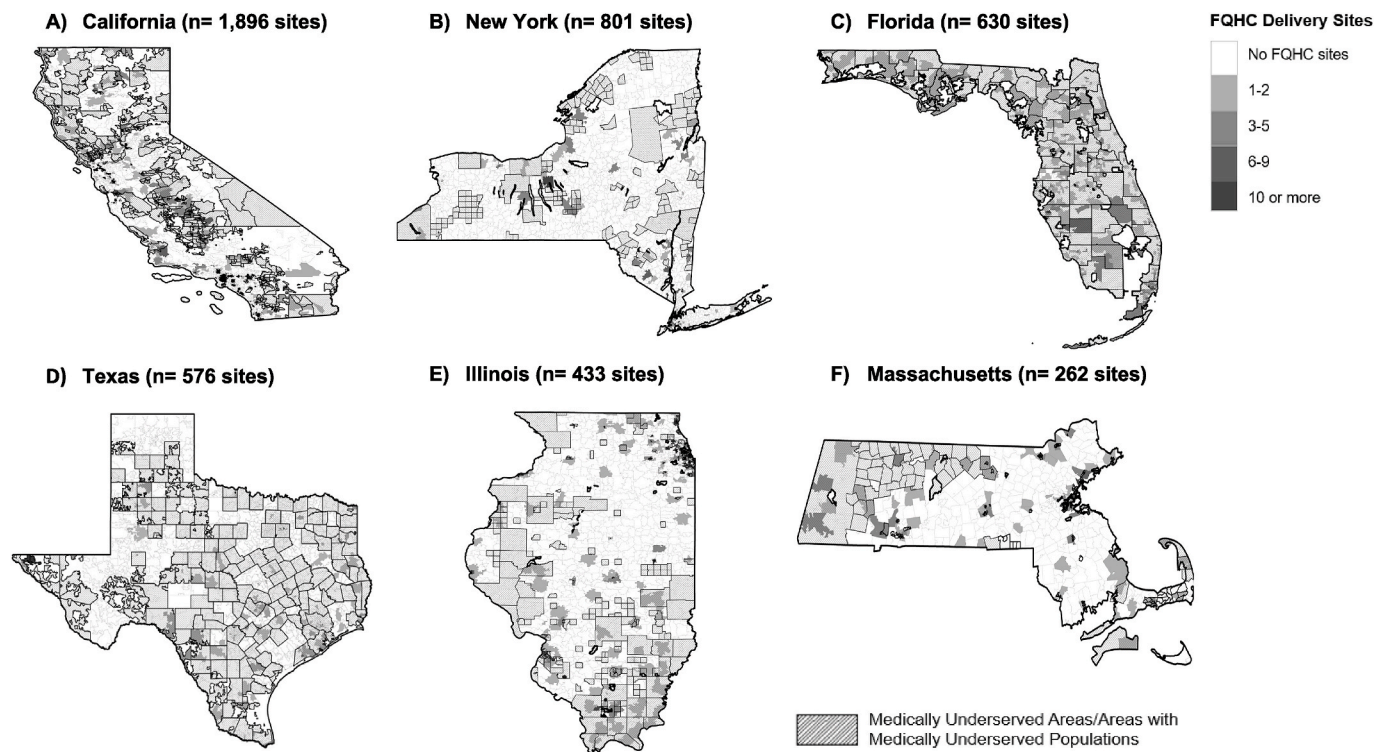


Fig. 2. Zip code-level maps showing the distribution of federally qualified health center (FQHC) delivery sites in six U.S. states. Black outlines with grey shaded areas represent medically underserved areas or areas serving underserved populations, as defined by HRSA.

were not statistically significant (Table 3). In contrast, we found a decrease in the prevalence of annual routine care visits with increasing zip-code level availability of FQHCs in metropolitan ($\beta = -0.148$; $p = 0.002$) and micropolitan areas ($\beta = -0.017$; $p = 0.946$) and small towns ($\beta = -0.385$; $p = 0.339$), although the relationships in micropolitan areas and small towns were not statistically significant.

Furthermore, among residents that had access to FQHCs within their zip codes, the likelihood of patients utilizing FQHCs for health services was significantly higher in worse HOLC-graded areas when compared to adjacent better HOLC-graded areas (B vs. A: $RR = 1.79$, 95%CI: 1.68–1.86; C vs. B: $RR = 1.68$, 95%CI: 1.64–1.71; and D vs. C: $RR = 1.24$, 95%CI: 1.21–1.27) (Fig. 3). We found significant variability across states with regards to how redlining scores were associated with FQHC utilization. For example, the likelihood for patients to utilize FQHCs was 41% (95%CI: 1.38–1.45), 42% (95%CI: 1.39–1.45) and 73% ($RR = 1.73$, 95%CI: 1.67 = 1.80) higher in D-graded zip codes compared to C-graded zip codes in California, Massachusetts and Florida, respectively. We found that the likelihood of FQHC utilization was 69% (95%CI: 0.30–0.32), 72% (95%CI: 0.27–0.29), and 78% (95%CI: 0.20 = 0.23) lower in D-graded zip codes compared to C-graded zip codes in Illinois, New York and Texas, respectively. We also observed a decrease in the prevalence of routine care visits with worsening HOLC graded areas ($\beta = -0.082$; $p = 0.750$).

4. Discussion

We investigated whether present-day availability of FQHCs and historic redlining was associated with higher utilization of health care services and its potential implications for improving inequities in health care access and disease prevention. We found that FQHC utilization rates were higher in zip codes with at least one FQHC site available compared to zip codes with no FQHC sites available. These associations were particularly evident in small towns, metropolitan areas, and worse HOLC-graded sections of urban areas when FQHC sites were available. However, we also found decreases in routine health visits at any health

clinic or facility in zip codes with FQHCs, potentially due to contextual factors associated with FQHC locations. Findings from this study underscore the value and potential benefits of FQHCs, and there are key implications for expanding access to FQHCs in key geographic areas (i.e., small towns, metropolitan areas and redlined sections of urban areas within MUAs): (1) expanding health care capacity to reach the medically underserved populations who are in greater need of medical treatments and preventive care; (2) reducing health inequities and outcomes through accessible health care system (i.e., FQHCs); and (3) maximizing the use of funding available to support expansion of FQHCs.

4.1. Associations by states, geographic areas, and FQHC availability

Across all six states, we observed that the likelihood of utilizing FQHCs in zip codes with at least one FQHC sites available were higher than their counterpart zip codes with no FQHC sites available. These relationships varied significantly across the six states and these findings remained partially true for health care use at FQHCs only. FQHCs are important because they have been linked to positive health outcomes and reduced disparities, and findings from this study can inform where new FQHCs should be located. Findings also suggest that additional outreach or strategies may be needed to ensure that populations living in MUAs get the routine, preventative care that they need, given other barriers outside of the healthcare system (for example, childcare, transportation, or workplace inflexibilities). Specific ways in which state-level interventions and policies may influence variation across states include, differences in: (1) state spending on public health and other social services, influencing variability in access and capacity of health care services in FQHCs (Bradley et al., 2016; Trust for America's Health and Robert Wood Johnson Foundation, 2014); (2) funding opportunities based on local initiatives—certain areas or community centers may be seeking for funding more actively than others to support their services and/or expansion of additional sites, which may influence differences in the availability of FQHCs; and (3) adaptation of Medicaid expansion and Medicaid eligibility (i.e., these policy changes may cover

Table 2

Associations between zip code-level availability of federally qualified health centers (FQHCs) and FQHC patients in six states, 2019.

	RR (95% CI) for FQHC visits ^b			
	Unadjusted models	P value	Adjusted models	P value
States				
California ^a	1.48 (1.45, 1.51)	<.001	3.94 (3.44, 4.51)	<.001
Florida ^b	5.26 (5.09, 5.44)	<.001	1.12 (0.93, 1.34)	0.251
Illinois ^b	4.30 (4.20, 4.39)	<.001	2.91 (1.64, 4.82)	<.001
Massachusetts ^c	10.2 (9.49, 10.9)	<.001	5.36 (4.13, 6.94)	<.001
New York ^d	3.35 (3.28, 3.43)	<.001	6.33 (4.03, 9.94)	<.001
Texas ^a	2.78 (2.71, 2.86)	<.001	5.24 (4.48, 6.14)	<.001
All 6 states ^e	3.56 (3.52, 3.59)	<.001	3.27 (2.27, 4.70)	<.001
FQHC availability^g				
Low: 1 FQHC sites ^a	2.33 (2.30, 2.35)	<.001	2.07 (1.86, 2.30)	<.001
Medium: 2–4 FQHC sites ^e	3.24 (3.20, 3.27)	<.001	5.48 (3.64, 8.25)	<.001
High: ≥5 FQHC sites ^e	5.33 (5.27, 5.38)	<.001	6.38 (5.30, 7.68)	<.001
Location of FQHCs				
Metropolitan areas ^f	4.36 (4.28, 4.44)	<.001	4.51 (3.18, 6.39)	<.001
Micropolitan areas ^f	2.64 (2.59, 2.70)	<.001	1.65 (1.40, 1.96)	<.001
Small towns ^a	4.62 (4.51, 4.73)	<.001	6.07 (5.18, 7.12)	<.001
Rural areas ^a	5.74 (5.58, 5.90)	<.001	2.07 (1.82, 2.35)	<.001

Abbreviations: FQHC, Federally Qualified Health Center; RR, rate ratio; CI, confidence intervals.

^a Models were adjusted for zip code-level non-US citizenship and household car ownership.

^b Models were adjusted for zip code-level lower educational attainment (i.e., less than high school education), non-US citizenship and household car ownership.

^c Models were adjusted for zip code-level non-US citizenship, unemployment, and household car ownership.

^d Models were adjusted for zip code-level unemployment, households receiving Supplemental Nutrition Assistance Program (SNAP), percent minority, and household car ownership.

^e Models were adjusted for zip code-level percent minority, non-US citizenship, and household car ownership.

^f Models were adjusted for zip code-level unemployment, percent minority, non-US citizenship, and household car ownership.

^g FQHC sites per zip code.

^h RRs were estimated based on e^h

more low-income Americans by increasing income eligibility from 133 percent to up to 138 percent of the federal poverty level), influencing insurance coverage and utilization of health care services (Trust for America's Health and Robert Wood Johnson Foundation, 2014).

We also found that certain socio-demographic and economic factors (i.e., non-U.S. citizenship, vehicle ownership, lower educational attainment, unemployment, households receiving SNAP benefits) may explain the differences in accessing and utilizing health care services (i.e., FQHCs and any health clinic/facility) across states. This was particularly evident when we observed that zip codes with at least five FQHC sites had 3.3 times higher likelihood of patients utilizing FQHCs compared to zip codes that did not have any FQHCs, suggesting that FQHCs can be impactful for ameliorating disparities in health care access for residents in MUAs. Potential explanations for differences across states may include differences in: (1) access to culturally and linguistically competent resources and services (i.e., translated materials,

Table 3

Associations between zip code-level availability of federally qualified health centers (FQHCs) and prevalence estimates of annual routine care visits in six states, 2019.

	β (SE) for prevalence estimates of routine care visits			
	Unadjusted models	P value	Adjusted models ^a	P value
States				
California	−0.259 (0.057)	<.001	−0.037 (0.052)	0.479
Florida	−1.844 (1.015)	0.091	−0.307 (0.659)	0.661
Illinois	−0.295 (0.097)	0.003	−0.107 (0.077)	0.167
Massachusetts	−0.092 (0.114)	0.425	0.174 (0.118)	0.148
New York	−0.139 (0.052)	0.009	−0.060 (0.050)	0.229
Texas	−0.607 (0.133)	<.001	−0.130 (0.115)	0.259
All 6 states	−0.220 (0.056)	<.001	−0.122 (0.046)	0.008
Location of FQHCs				
Metropolitan areas	−0.157 (0.062)	0.012	−0.148 (0.046)	0.002
Micropolitan areas	−0.598 (0.342)	0.084	−0.017 (0.250)	0.946
Small towns	−0.997 (0.462)	0.035	−0.385 (0.399)	0.339
Rural areas	−1.364 (0.812)	0.098	0.269 (0.646)	0.679

Abbreviations: FQHC, Federally Qualified Health Center; RR, rate ratio; CI, confidence intervals.

^a Models for each state were adjusted for zip code-level proportion of minority population (i.e., Black, American Indian, Pacific Islander, other and two or more races), lower educational attainment (i.e., less than high school education), non-US citizenship, proportion of homes with no computer device, proportion of home receiving Supplemental Nutrition Assistance Program, proportion of homes owning a vehicle, unemployment, total population.

community-based organizations) and established trust between community members; or (2) access to public transportation, which may be more accessible in large cities compared to rural areas or small towns (Andersen et al., 2002; Berdahl et al., 2007; Cordasco et al., 2011). We may also further speculate that access to transportation and citizenship status (i.e., having less access to drivers' license or health insurance due to immigration status) may contribute to fewer routine care visits (Arcury et al., 2015; Wallace et al., 2005; Wolfe et al., 2020). Moreover, households participating in federal nutrition assistance programs (e.g., SNAP) are more likely to be in greater need of health care services, partly as they may be more prone to chronic diseases and infections as a result of diet and life events (e.g., job loss during COVID-19, discrimination, family deaths) increasing life burdens and stress levels, leading to worse health outcomes (Abrams et al., 2020; Ciesielski et al., 2021; Peltz and Garg, 2019). Such disparities in socio-economic and demographic factors (e.g., fewer access to resources, higher rates of poverty, lack of health insurance, higher proportion of minority population) may explain our findings on fewer routine care visits in the presence of FQHCs. Mechanisms through which contextual factors and social forces play a role on health services utilization warrants further investigation on a neighborhood-specific basis.

4.2. Associations by neighborhood disparities

This study found that zip codes designated with worse HOLC-grades had higher likelihood of patients utilizing FQHCs compared to zip codes with better HOLC-grade designations, and we found significant variation across six states. Communities in historically redlined and racialized neighborhoods are more affected by these disparities because of differences in socioeconomic forces (i.e., access to quality education, jobs, housing, safety, social networks) and environmental risk factors (i.e., poor housing conditions, living nearby polluting facilities, oil and gas wells) that influence health needs (Charles, 2003; Cushing et al., 2022; Gaskin et al., 2012a; Gonzalez et al., 2022; Lane et al., 2022; Retzlaff, 2020). School-aged children in worse-graded neighborhoods may have lower access to quality education and experience long-term effects with worsened early-adult outcomes, including lower educational attainments, labor market earning and higher incarceration rates (Lukes and

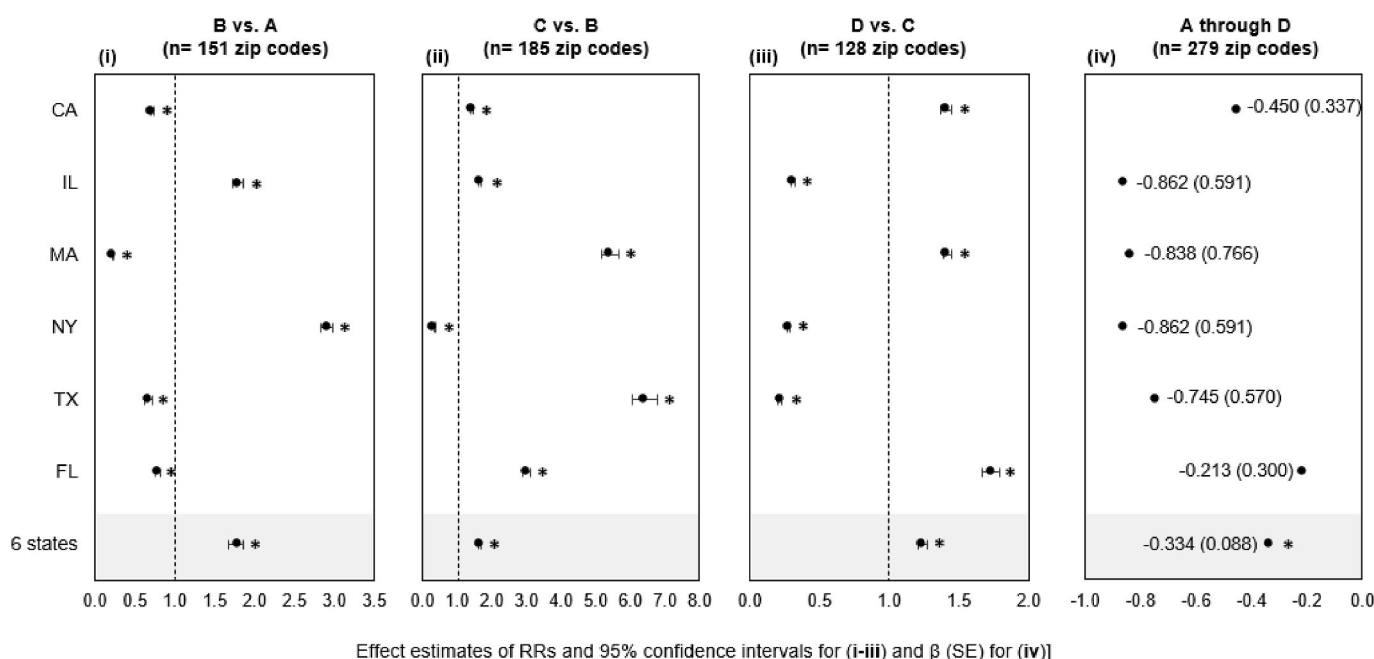


Fig. 3. Poisson (i-iii, rate ratio estimates for FQHC patients) and multivariate regression associations (iv, β estimates for routine care visits) with the 1938 Home Owners' Loan Corporation (HOLC) grades in FQHC available zip codes across six states, 2019. Error bars represent 95% confidence intervals and asterisks (*) represent all effect estimates that were statistically significant with $p < 0.05$. All models were adjusted for unemployment, vehicle ownership and total population.

Cleveland, 2021). Our findings showing variation across states may also reflect the disparities of persistent disinvestment versus gentrification and urban development (i.e., C and D graded areas), and potential mediation effects via racialized economic segregation (Metzger and Webber, 2018; Nelson et al., 2019; Rothstein, 2017).

Additionally, the geographic differences resulting from historical housing policies and residential segregation may also reflect present-day variation in primary care physicians, Medicaid provider networks, and the extent of collaborations among health and social services influencing health care access (Department of Health and Human Services, 2014; Goodman et al., 2010). This is particularly important to note given that, similar to the findings in this study, a recent study found decreased routine health care visits (at any hospital, clinic, or health center) among residents living in redlined neighborhoods compared to those residing in a non-redlined neighborhood (Motairek et al., 2022). These indirect mechanisms of structural inequality may indicate greater need for health care services capable of providing coordinated, holistic, and culturally appropriate care (i.e., FQHCs) in historically marginalized neighborhoods. It is therefore important to consider that for provisioning health care to marginalized and underserved population, availability of FQHCs may be impactful for reducing health inequities particularly in worse HOLC graded areas and these areas should be considered for future siting of FQHCs.

Furthermore, there might be indirect mechanisms for which social and built environmental factors may influence utilization of health care services. Several studies in the U.S. and worldwide have reported associations between greater presence of dis-amenities (i.e., industrial facilities, landfills, gas wells) (Brender et al., 2011; Casey et al., 2019; Hajat et al., 2015; Lee et al., 2021; Tessum et al., 2021; Vrijheid, 2000) and detrimental social structures (i.e., crime and poverty rates, racial segregation) (Benms et al., 2020; Jacoby et al., 2018; Mccarthy et al., 2015) and increased adverse health outcomes in socioeconomically and racially marginalized neighborhoods. Greater availability of amenities (i.e., access to green space, recreational resources) and healthy foods in neighborhoods may lower the need to seek health care services due to improved health, including lowered blood pressure and fewer adverse cardiovascular outcomes (Kondo et al., 2018; Krukowski et al., 2010; Lachowycz and Jones, 2011; Twohig-Bennett and Jones, 2018; Vanaken

and Danckaerts, 2018). Greater exposure to dis-amenities (Lee et al., 2021; Vrijheid, 2000) and detrimental social structures (e.g., racial segregation, policing, criminalization of poverty) (Benms et al., 2020; Jacoby et al., 2018; Mccarthy et al., 2015) may increase adverse health outcomes, thereby increasing the need for utilizing health care services. These social and environmental forces may increase health care needs, which may not always align with availability of health care services and suitable environments to access health care facilities.

4.3. Public health implications

Our study provides evidence for targeting future FQHC expansion by identifying geographic areas and zip code-level characteristics associated with FQHC seeking patients. Establishing and expanding FQHCs in geographic areas where need is the greatest (including redlined communities, small towns, and metropolitan areas) may maximize FQHC impact, including FQHC impact on ameliorating health inequities. Within existing FQHCs, increasing operational hours (i.e., evenings, weekends), providing transportation to appointments, offering home visit services, and enhancing preventive care services (e.g., FQHC-based farmers' markets) (Freedman et al., 2012, 2013, 2014) may aid in further improving access to FQHC health care services (Knight, 2011).

4.4. Limitations

This study has some limitations. First, there could be misalignment between zip code-level sociodemographic information and FQHC patient volume, since some patients may not have used FQHCs located in their own residences' zip codes (e.g., no FQHCs was available near their residences). However, this may reflect the reality of how communities navigate and utilize health care services and micro-level analyses should be considered in future studies. Additionally, to reduce potential for exposure misclassification, we selected unexposed areas ensuring that no nearby FQHC sites (i.e., within eight miles) would be located. We also recognize that unmeasured potential confounders such as environmental factors (i.e., housing conditions), behavioral risk factors (e.g., smoking, diet, physical activity) and pre-existing health conditions (e.g., stress, co-morbidities, family medical history) may have introduced

confounding bias. To address this concern, we controlled for potential confounders primarily known to associate with health care utilization, including household car ownership, non-U.S. citizenship, race/ethnicity, low educational attainment and other socio-economic factors in the statistical analysis in addition to the matching of sites by MUAs, as described above.

Furthermore, our analyses were based on area-level data, rather than individual-level data, due to data availability, thus findings from this study may be prone to the ecologic fallacy. Similarly, because of data unavailability (e.g., FQHC utilization data available at the facility-level, whereas prevalence of routine care visit estimates available at the patient-level), our findings should be interpreted according to the limitations presented with data. Nevertheless, this study provides valuable baseline knowledge to generate hypotheses to be tested in future studies with more robust epidemiological study design, data and/or analysis covering larger geographic areas (i.e., nationwide analyses). Lastly, this study represents findings based on six U.S. states and limited to including FQHC zip codes that met the set criteria (i.e., being in MUAs), accounting for nearly half of the FQHC sites in these six states; thus, caution should be exercised when generalizing findings from this study.

5. Conclusions

This study found that residents of MUAs were more likely to use FQHCs if there was an FQHC in their zip code. This relationship was stronger among MUA residents living in small towns and metropolitan areas, and worse in HOLC-graded sections of urban areas. However, we found that these relationships did not remain true for routine care visits at any other health clinic or facility. This suggests that availability of FQHCs may influence FQHC care seeking behavior for people in MUAs if there was one available, and the FQHC system can aid in mitigating existing barriers to important health services (e.g., preventive care) that may be particularly impactful in small towns, metropolitan areas, and redlined areas. This hypothesis could be further tested in future research with longitudinal studies evaluating whether geographic expansion of FQHCs (i.e., pre- and post) improves access to critical health services (e.g., preventive care, chronic disease management) and health outcomes among FQHC patients. Because FQHCs can provide high quality, culturally competent, cost-effective access to important primary care, behavioral health, and can enable services that uniquely benefit low-income and marginalized patient populations, improving availability of FQHCs may be an important mechanism for improving health care access through the FQHC system and reducing subsequent inequities for these underserved groups.

Credit author statement

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Data availability

All data used in this manuscript is publicly available.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.socscimed.2023.116009>.

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